'Environmental Management in Development,' World Bank Discussion Papers 80, Washington D.C., 1990.

21. Economists use the term 'externality' to refer to spillover effects that are not accounted for by the marketplace. For example, air or water pollutants that are by products of an industrial process are environmental externalities. Firms do not have the incentive to reduce or eliminate such pollutants unless economic penalites such as emissions taxes are applied, or unless the firms are required to do so by regulation. See William D. Nordhaus, ***"The Ecology of Markets,"*** Proc. Natl. Acad. Sci. USA, Vol. 89, pp. 843-850, February 1992.
22. Human capital in this sense refers to 'knowledge' or technological capital. This notion of substitutability has been called the criterion of 'weak sustainability'. See Herman E. Daly and John B. Cobb, ***For the Common Good: Redirecting the Economy Toward Community, the Environment, and a Sustainable Future,*** Beacon Press, USA, Boston, 1989.
23. Thomas L Friedman, Hot, Flat, and Crowded, Penguin Books, Great Britain, London, 2009.
24. This has been called the criterion of ***'strong sustainability',*** Daly & Cobb, op. cit.
25. Even though advanced industrial societies are becoming increasingly efficient in their use of materials, a phenomenon known as ***"demate- rialisation,"*** in the eyes of some, greater industrial efficiency by itself is not a sufficient response to environmental problems. Indeed the absolute quantities of materials consumed and wastes produced are increasing; they are just not increasing as fast as GNP.
26. Moreover, within each idea, technologists will likely be confronted with a variety of difficult tradeoffs. For instance, there are typically many environmental tradeoffs associated with the use of a specific material. As an illustration, the new class of high temperature superconductors, which potentially offer vast improvement in power transmission efficiency and have other promising applications, are quite toxic; the best of the superconductors is based on thallium, a highly toxic heavy metal. The fact that products that use toxic materials can perform socially useful functions, or even have comparative environmental benefits, underscores the need for a flexible approach to environmental questions. For additional discussion of this issue see 'Green Products By Design', op. cit., footnote 6.
27. See U.S. Congress, Office of Technology Assessment, ***Beneath the Bottom Line: Agricultural Approaches To Reduce Agrichemical Contamination of Groundwater,*** OTA-F-418 (Washington, DC: U.S. Government Printing Office, November 1990).
28. But to encourage decision making on a system-wide basis, utilities need to be allowed to benefit financially from investments in efficient end-use equipment. Recent changes in regulatory frameworks have played a key role in moving utilities in this direction. See U.S. Congress, Office of Technology Assessment, ***Energy Technology Choices: Shaping Our Future,*** OTA-E-493, (Washington, DC: U.S. Government Printing Office, July 1991).
29. Walter Stahel, The Product-Life Institute, Geneva, Switzerland; Personal Communication. For more on this idea see Orio Giarini and Walter Stahel,***"The Limits to Certainty: Facing Risks in the New Service Economy,"*** (Boston, MA: Kluwer Academic Publishers, 1989).
30. In an ultimate sense, a true systems philosophy involves consideration of many different perspectives, and not simply an economic or technological perspective. Humanity is now confronted with a series of global problems such as rapid growth and migration of populations, crushing poverty, intractable religious and ethnic conflicts, and widescale ecological damage – that in one way or another are linked together. "None of these problems can be fully addressed without considering all the others." See ***"The Most Vital Challenge,"***